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EE1030/EE2100: Matrix Theory

Lecture 20: Matrix Multiplication

Topics :

1. Matrix Multiplication
 2. Matrix Vector Product
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Hermitian and Skew Hermitian Matrices (Complex Matrices)

a) if $A \in \mathbb{C}^{n \times n}$ then $A^H \leftarrow$ Hermitian Transpose is a matrix whose entries are

$$(A^H)_{ij} = \overline{(A)_{ji}} = \overline{a_{ji}}$$

b) A matrix $A \in \mathbb{C}^{n \times n}$ is Hermitian if $A^H = A$.

↓
diagonal entries $\overline{a_{ii}} = a_{ii} \Rightarrow$ Real entries

c) A matrix $A \in \mathbb{C}^{n \times n}$ is skew Hermitian if $A^H = -A$.

↓
diagonal entries $\overline{a_{ii}} = -a_{ii} \Rightarrow$ Purely imaginary.

• Set of all Hermitian matrices of vector space $\mathbb{C}^{n \times n}$ defined over $\mathbb{C} \rightarrow$ Subspace? (No)
↓
 $\exists A \notin$ Subspace

Result: Any square matrix $A \in \mathbb{C}^{n \times n}$ can be written as sum of Hermitian & skew Hermitian matrix

$$\begin{bmatrix} 1 & 1-i \\ 1+i & i \end{bmatrix} \sim \begin{bmatrix} 0 & -i \\ i & i \end{bmatrix}$$

Matrix Multiplication (Commutative and Associative Properties)

Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times q}$. The product AB is defined if $p=n$ & is given

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

$$\rightarrow \begin{bmatrix} A_{11} & \dots & A_{1n} \\ \vdots & & \vdots \\ A_{m1} & \dots & A_{mn} \end{bmatrix} \begin{bmatrix} B_{11} & \dots & B_{1q} \\ \vdots & & \vdots \\ B_{n1} & & B_{nq} \end{bmatrix} =$$

if A is considered as a collection of row vectors & B as a collection of col vec

$$\begin{bmatrix} \vec{a}_1^T \\ \vdots \\ \vec{a}_m^T \end{bmatrix} [\vec{b}_1 \dots \vec{b}_q] \quad \text{then } (AB)_{ij} = \underbrace{\vec{a}_i^T \vec{b}_j}_{\text{inner product.}}$$

from earlier course a) $AB \neq BA \quad \forall A, B$

$$b) (AB)C = A(BC)$$

$$c) (AB)^T = B^T A^T \quad (\text{to be proved later})$$

Matrix Vector Product - Form II

If $\vec{v} \in \mathbb{R}^m$ and $A \in \mathbb{R}^{m \times n}$, then $\vec{v}^T A \in \mathbb{R}^{1 \times n}$ and $(\vec{v}^T A) = \vec{y}^T$ where $y_i = \vec{v}^T \vec{a}_i$
 $\vec{v}^T$ ← Row vector
 $\vec{a}_i$ ← Column vector of A

$$[\vec{v}_1 \dots \dots \vec{v}_m] \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix}$$

$$\vec{v}^T A = [v_1 a_{11} + \dots + v_m a_{m1}, \dots, v_1 a_{1n} + \dots + v_m a_{mn}]$$

$$= \text{linear comb of row vectors of } A = \sum_{i=1}^m v_i \begin{pmatrix} a_{i1} \\ \vdots \\ a_{in} \end{pmatrix}$$

row vector of A

if $A = \begin{bmatrix} \vec{a}_1^T \\ \vdots \\ \vec{a}_m^T \end{bmatrix}$, then $\text{Span} \{ \vec{a}_1^T, \dots, \vec{a}_m^T \} \in \mathbb{R}^{1 \times n}$ = Set of all possible linear comb of $\vec{a}_i$
 $= \{ \vec{v}^T A \}$ where $\vec{v} \in \mathbb{R}^m$
 $\vec{v} \in \mathbb{R}^m$ → every vector in $\mathbb{R}^m$

Row Space of A : $\text{Row}(A) \rightarrow$ any representative rep.?

Pre and Post Multiplication of a Matrix with Diagonal Matrices

Let $A \in \mathbb{R}^{m \times n}$ & $D = \text{Diag}(d_{11}, \dots, d_{mm}) \Leftrightarrow$ Diag matrix of size (m, m)

$$DA = \begin{bmatrix} d_{11} \vec{a}_1^T \\ \vdots \\ d_{mm} \vec{a}_m^T \end{bmatrix} \quad \text{where } \vec{a}_i^T \rightarrow i^{\text{th}} \text{ row vector of } A$$

Let $A \in \mathbb{R}^{m \times n}$ & $E = \text{Diag}(e_{11}, \dots, e_{nn}) \Leftrightarrow$ Diag matrix of size (n, n)

$$AE = [e_{11} \vec{a}_1 \quad \dots \quad e_{nn} \vec{a}_n] \quad \text{where } \vec{a}_i \rightarrow i^{\text{th}} \text{ column vector of } A.$$